Modern MONOPOLY
515 Team Project Sienna&Polly

Section1: System Mission & Boundary

- What does the system do?

The system is a hybrid board game that automatically tracks piece positions, player assets, and game events by combining a physical board with embedded sensing and digital interfaces. It coordinates gameplay in real time and reduces manual bookkeeping.

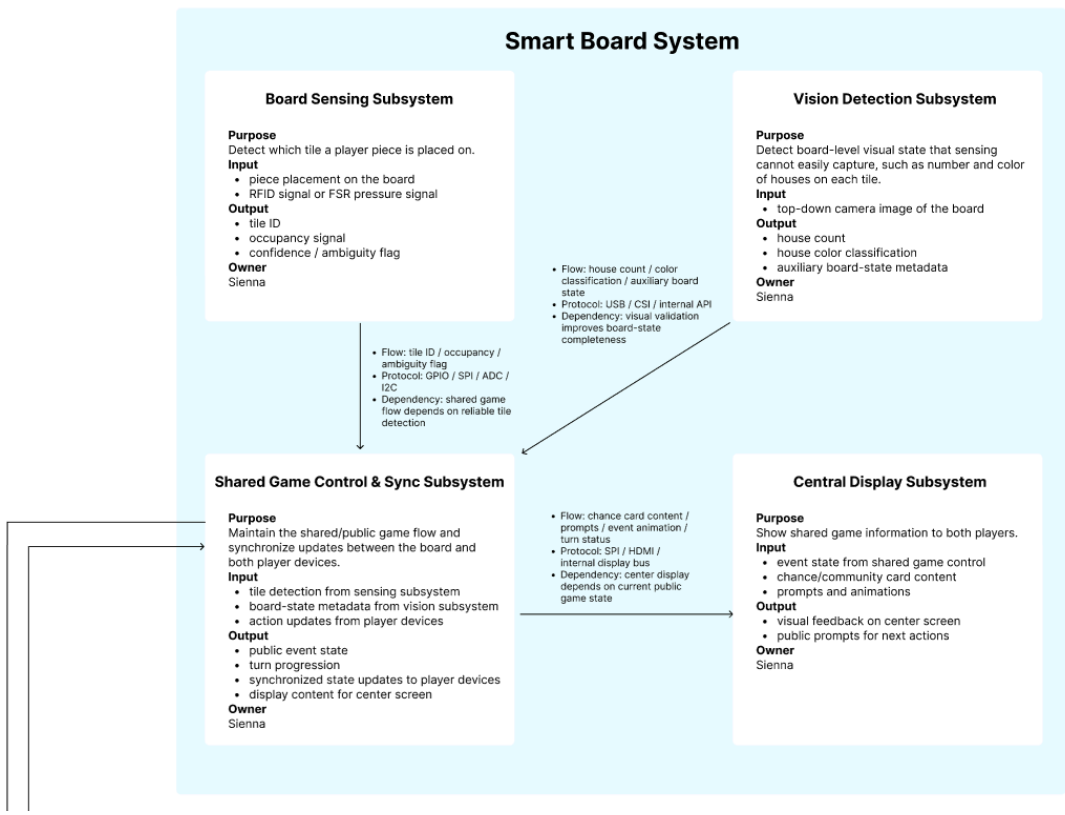
- Who is it for?

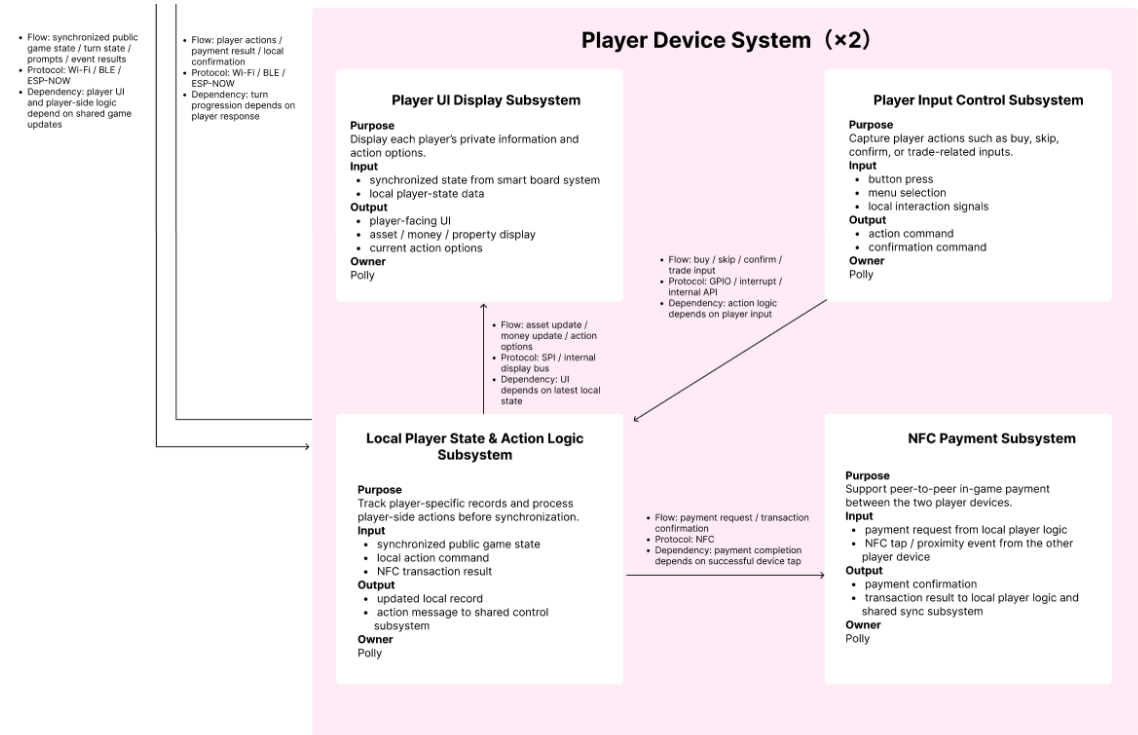
It is designed for two players who want a physical, face-to-face board game experience with reduced complexity and enhanced interaction through digital assistance.

- What hardware, software, and data does it use?

The system uses a smart board with RFID/FSR sensing, an optional camera, and a central display, along with two ESP32-based player devices with screens and NFC; software is distributed across devices to process sensing data, manage player state, and synchronize game flow using position, visual, and interaction data.

Section2: Functional Architecture Diagram





Feedback loop:

Player moves a piece on the board → Board Sensing Subsystem detects the tile → Shared Game Control & Sync updates the public state → Central Display shows the event or prompt → Player responds on the Player Device → Local Player State & Action Logic sends the action result back to the Shared Game Control & Sync subsystem → the system advances to the next game state.

Integrated system outcome

By combining board sensing, shared game coordination, central event display, and private player devices, the integrated system creates a synchronized hybrid board game experience that neither the board nor the player devices could produce independently.

Section3: Risk Reduction Prototypes

Field	Board Sensing	System Synchronization	NFC Payment Interaction
Name + Owner	Tile Detection Prototype Sienna	State Synchronization Prototype Sienna&Polly	NFC Payment Prototype Polly
Assumption being validated	The system can correctly detect which tile a piece is on without confusion at tile boundaries	All devices can maintain a consistent game state with low latency	NFC interaction can reliably complete payments without state errors
Inputs -> Process -> Outputs	Inputs: piece placement across tiles Process: repeated placement tests near boundaries Outputs: detection accuracy, boundary error rate	Inputs: simulated game events and player actions Process: send updates across devices Outputs: sync latency, consistency rate	Inputs: payment requests and NFC tap Process: simulate transactions Outputs: success rate, transaction latency

Field	Game Logic	Touchscreen UI Interaction
Name + Owner	Distributed Game Logic Consistency Prototype Sienna&Polly	Touchscreen UI Interaction Prototype Sienna
Assumption being validated	Distributed logic across board and devices can stay consistent without conflicts	Players can understand and operate the touchscreen interface efficiently
Inputs -> Process -> Outputs	Inputs: player actions and tile events Process: run test scenarios across devices Outputs: consistency rate, conflict rate	Inputs: UI interaction tasks Process: user testing Outputs: task time, error rate, usability feedback